

Books Reconciliation Report

Trip Fee Payment Verification & Follow-Up Analysis

1. What This Report Is About

This report checks whether the money collected for a class trip matches the amount that was actually expected. A Python script was written to go through every payment record, figure out which ones were meant for the trip, add them up, and compare that total against the amount that was manually verified beforehand. Any payment that doesn't clearly look like a trip payment is flagged separately so it can be followed up on.

2. How the Script Works (In Simple English)

Step 1 — Load the payment data

The script uses the pandas library to read a CSV file (payments.csv) containing every payment that was received, along with the sender's name, the amount in PKR, and a memo (a short note describing what the payment was for).

Step 2 — Set the verified total

A fixed number, Rs. 6000, is entered as the “verified total” — this is the amount that was manually confirmed as the correct amount that should have been collected for the trip.

Step 3 — Define what counts as a “trip payment”

A list of acceptable memo texts is created (such as “trip”, “trip fee”, “class trip”, “classtrip”, “trip2026”, and the short form “trp”). Any payment whose memo matches one of these (regardless of capital/small letters or extra spaces) is treated as a trip payment.

Step 4 — Fix a spelling shortcut

One sender had entered their name in short form (“A. Ali”). The script replaces this with the full name “Ahmed Ali” so the record is clean and consistent.

Step 5 — Apply a smart rule for blank memos

Some payments had no memo written at all (blank / NaN). Instead of ignoring these, the script applies a reasonable assumption: if the memo is empty AND the amount is exactly Rs. 500 (the known trip fee per student), it is still counted as a trip payment. This prevents genuine trip payments from being missed just because someone forgot to write a note.

Step 6 — Add everything up and compare

All payments identified as trip payments are added together to get the “calculated total.” This is then compared with the Rs. 6000 verified total. The difference between the two tells us whether any money is missing or unaccounted for.

Step 7 — Identify who still needs follow-up

Any payment that did NOT qualify as a trip payment (i.e. it doesn't match the accepted memo list and isn't a blank Rs. 500 entry) is separated out into a “follow-up” list, so that person can be asked directly what their

payment was for.

Step 8 — Print a clear summary

Finally, the script prints a clean, readable report showing the verified total, the calculated total, the difference, the list of trip payments, and the list of payments needing follow-up.

3. Output — Verified Against the Script's Logic

The script was run on the uploaded payments.csv (12 records). Below is what it produced, checked line by line against the rules in Step 3–5 above.

Metric	Value	Check
Verified Total (manual)	Rs. 6,000	Given
Calculated Total (script)	Rs. 5,500	11 trip payments x Rs. 500
Difference	Rs. 500	6000 - 5500 = 500 (correct)
Unmatched Payment	Zain Malik - Rs. 500 - Memo: "Gift"	Correctly excluded

Why the math checks out:

The script identified 11 payments as trip-related (matched either by memo text or the blank-memo + Rs. 500 rule). $11 \times \text{Rs. } 500 = \text{Rs. } 5,500$. Subtracting this from the verified total of Rs. 6,000 leaves exactly Rs. 500 unaccounted for — which matches the one payment the script correctly excluded (Zain Malik's Rs. 500 payment marked “Gift”, which is not a trip-related memo and was not blank). The logic and the arithmetic are consistent with each other.

Trip Payments Identified (11 records)

Sender Name	Amount (PKR)	Memo
Ali Raza	500	Trip Fee
Sara Khan	500	trip
Ahmed Ali	500	Class Trip
Ahmed Ali	500	trp
Fatima Noor	500	(blank)
Hassan Ahmed	500	Trip2026
M. Khan	500	Trip Fee
Ayesha Tariq	500	trip fee
Umer Farooq	500	Trip
Maryam Iqbal	500	ClassTrip
Bilal Khan	500	(blank)

Payment Requiring Follow-Up (1 record)

Sender Name	Amount (PKR)	Memo	Reason Flagged
Zain Malik	500	Gift	Memo does not match trip keywords

4. Final Summary

Rs. 6,000	Rs. 5,500	Rs. 500	Zain Malik
Verified Total	Calculated Total	Difference	Unmatched Person

Conclusion: The script's logic and output are verified as correct. Rs. 500 remains unaccounted for, and it traces directly to Zain Malik's payment (memo: "Gift"), which does not fit any trip-related pattern. This person should be contacted to confirm whether the payment was actually intended for the trip or was genuinely a separate gift.

5. Reconciliation Checklist

- ✓ Manually verified total provided.
- ✓ Original payment record used.
- ✓ Interpretation rules applied.
- ✓ Python script written and executed.
- ✓ Total calculated and compared.
- ✓ Difference identified.
- ✓ Follow-up person identified.
- ✓ Reconciliation complete.